Zahra Moslemi

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Research Focus

Ph.D. candidate in Statistics and Machine Learning with research across **LLMs/agents, multimodal and generative modeling, Bayesian ML/uncertainty, reinforcement learning, and robust representation learning**. Three Research Scientist internships and peer-reviewed work at TMLR, NeurIPS workshops, and AAAI workshops span both foundational methodology and applied ML systems.

Education

University of California, Irvine 2023–Present
Ph.D. in Statistics and Machine Learning (GPA: 3.86 / 4), ***Expected January 2027***

University of California, Irvine 2021–2023
M.Sc. in Statistics and Machine Learning (GPA: 3.86 / 4)

- **Selected coursework:** Machine Learning, Data Mining, High-Dimensional Computing, Bayesian Inference, Random Processes, Design under Uncertainty, Deep Generative Models, Data Structures and Algorithms.
- **Thesis:** Scalable Bayesian Transformers with GP-inspired function priors and attention-guided selective uncertainty.

Sharif University of Technology 2016–2020
B.Sc. in Industrial Engineering, ***ranked 4th of 200*** *(GPA: 3.94 / 4.00)*

Research & Industry Experience

Research Scientist Intern – ClearNote San Mateo, CA, Summer 2026

- Developed **FragJEPA**, a biology-informed **Vision Transformer** for sparse pancreatic-cancer cfDNA fragmentomics using multi-view joint-embedding prediction for robust molecular representation learning.
- Developed **conformal and Bayesian inference** methods for uncertainty-aware prediction, integrating coverage-controlled prediction sets with posterior uncertainty estimation.

Research Scientist Intern – Accenture, Center for Advanced AI | *T-Mobile collaboration* Boston, Summer 2025

- Developed and evaluated **AutoGen-based multi-agent systems** that reason, plan, and act autonomously through tool use, retrieval-augmented reasoning, goal management, and feedback-driven learning.
- Implemented a **planner–evaluator** pattern with DSPy few-shot learning and rubric-guided judges, producing interpretable reasoning traces and reproducible evaluations across scenarios.
- Built a **ReAct + Chainlit** system and integrated Azure AI Foundry Agent Service for human-in-the-loop review, telemetry, and controlled rollouts; applied **reinforcement-fine-tuning-updated** planning priors/selector weights and audited robustness under distribution shift. **Led to two AAAI 2026 workshop papers.**

Research Scientist Intern – ClearNote San Mateo, CA, Summer 2024

- Built a **DDPM synthetic-data pipeline** followed by **Perceiver IO** fine-tuning to improve minority-class AUCPR under class imbalance, domain shift, and data scarcity.
- Prototyped **graph-aware fusion** for calibrated and robust risk prediction; added uncertainty quantification, ECE, ablations, and reproducible evaluation.
- Productionized experiments with **Nextflow**, dataset versioning, and AWS/GCP/Azure compute for scalable, observable, compliance-ready experimentation.

Selected Publications and Preprints

Z. Moslemi, Z. Liang, B. Shahbaba. *SCOPE Before You Sample: Conformal Scores Guide Posterior Exploration for Fast and Calibrated Bayesian Predictive Inference*. arXiv preprint, 2026; **under review at AISTATS 2027**. Score-guided posterior inference combines MCMC and variational approximation with conformal calibration; 12–15$\times$ faster than full HMC on the reported real-data benchmarks.

Z. Moslemi, Y. Meng, S. Lan, B. Shahbaba. *Scaling Up Bayesian Neural Networks with Neural Networks*. TMLR, 2024. Calibration–emulation–sampling for efficient Bayesian inference and uncertainty quantification.

Z. Moslemi, Z. Liang, N. J. Fortin, B. Shahbaba. *Multi-Graph Meta-Transformer: An Interpretable Framework for Cross-Graph Functional Alignment in Neural Decoding*. NeurIPS 2025 AI4Science Workshop (Spotlight); Best Poster Runner-up. Interpretable alignment across heterogeneous neural graphs.

Z. Moslemi, Z. Liang, N. J. Fortin, B. Shahbaba. *Multi-Graph Meta-Transformer (MGMT)*. NeurIPS 2025 NPGML Workshop (Spotlight). Attention-selected supernodes and meta-graph reasoning across heterogeneous graphs. Extended preprint: Heterogeneous Graph Alignment for Joint Reasoning and Interpretability, 2026.

Z. Moslemi, K. Koneru, Y.-T. Lee, S. Kumar, R. Radhakrishnan. *POLARIS: Typed Planning and Governed Execution for Agentic AI in Back-Office Automation*. AAAI 2026 Workshop on Agentic AI Benchmarks and Applications for Enterprise Tasks. Typed planning, validated execution, and auditable agent workflows.

Y.-T. Lee, K. Koneru, **Z. Moslemi**, S. Kumar, R. Radhakrishnan. *AEMA: Verifiable Evaluation Framework for Trustworthy and Controlled Agentic LLM Systems*. AAAI 2026 Workshop on Trust and Control in Agentic AI. Reproducible multi-agent evaluation with human oversight.

Z. Liang, **Z. Moslemi**, B. Shahbaba. *Neuroscience-Inspired Reinforcement Learning: From Episodic Control to Dynamic Memory Graphs*. Under review, Nature Machine Intelligence, 2026. Graph-based episodic memory and adaptive arbitration for reinforcement learning.

Selected Research Projects

Agentic Time-Series Foundation Models *(Google Research Collaboration)* 2026–Present

- Orchestrating time-series foundation models and reasoning LLMs for rare transitions in sparse, irregular longitudinal data, with uncertainty-aware model arbitration.
- Benchmarked Chronos, TimesFM, Moirai, and Toto across horizons under missingness, distribution shift, and rare events.

Continual, Multimodal & Edge Learning *(NSF $2.4M + NIH/NCI $1.2M)* 2023–Present

- Developed continual-learning, self-supervised multimodal sequence modeling, and domain-adaptation methods for irregular data, subject heterogeneity, and edge constraints.

Skills

Research Areas: LLMs/agents, multimodal learning, generative modeling, reinforcement learning, Bayesian ML, uncertainty quantification, robustness, domain adaptation
Programming: Python, SQL, KQL, MATLAB
Frameworks: PyTorch, Hugging Face, TensorFlow, Scikit-learn, GPyTorch, NumPy, Pandas, SciPy, Matplotlib
Systems: Azure, AWS, GCP, Azure AI Foundry, FastAPI, Nextflow

Selected Honors

NeurIPS 2025 Best Poster Runner-up, AI4Science.

NeurIPS 2025 Best Poster Award, New Perspectives in Graph Machine Learning.

ICML 2026 Gold Reviewer, recognized among top reviewers.

Henry Samueli Endowed Fellowship, University of California, Irvine, 2024.

Professional Activities

Data Science Mentor – COSMOS (NIH-funded), UCI School of ICS Summer 2022–2023

- Designed and taught ML/LLM modules covering embeddings, prompting, evaluation, statistical learning, and reproducible experimentation; mentored end-to-end AI projects with error analysis and responsible-AI practices.

Mentor & Academic Guide – SoCal Data Science / UCI Summer 2022–2023

- Guided applied LLM projects spanning RAG, lightweight fine-tuning, calibration, robustness, ablations, fairness, privacy, and reproducible MLOps.

Selected Training

Generative AI with Large Language Models – DeepLearning.AI; Operationalizing LLMs on Azure – Duke University.